Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 12

Full Marks: 30 *Deadline: 5:30 PM, 07-10-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a zip file which will contain all the .dat, .gp, and .pdf files of the task in their respective folders.
3) Each Gnuplot script file should be placed in its respective task folder (e.g., Task1, Task2..).
4) Your zip file should be named as <Roll No>_<Name>_<Lab Activity No>.
5) Example: B24CS055_Shareen_10.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1) To learn the fundamentals of Gnuplot scripting for creating structured 2D and 3D plots and visualizations.
2) To practice generating clustered histograms, heatmap and 3D map using real and synthetic datasets.
3) To develop skills in customizing plots with titles, labels, legends, styles, and color schemes for clarity.
4) To produce publication-ready outputs in PDF format with professional formatting and readability.

Task1: Plot a simple histogram to visualize the average monthly temperatures of a city using Gnuplot. **[10 Marks]**

You are provided with a file named avg_temperature.dat containing the average temperature (°C) for each month of the year.

File format:

Month	Temperature
Jan	14
Feb	16
Mar	21
Apr	26
May	30
Jun	33
Jul	31
Aug	30
Sep	29
Oct	25
Nov	20
Dec	16

Column 1 → Month (categorical x-axis)
Column 2 → Average temperature (°C)

Tasks

A. Data Preparation
   a. Ensure the data is in two columns: Month and Temperature.
   b. Save it as avg_temperature.dat.

B. Plot a Simple Histogram
   Use Gnuplot to generate a simple vertical bar (histogram) representing average monthly temperatures.

C. Plot Specifications:
    a. Output file: avg_temperature_histogram.pdf
    b. Terminal: pdfcairo
    c. Page size: 18 cm × 12 cm
    d. Font: "Times New Roman", 12
    e. Plot type: Simple histogram

D. Style Requirements:
    a. Histogram style:data histograms
    b. Fill style: solid (70% opacity)
    c. Box width: 0.7
    d. Border: visible

E. Axis & Labels:
    a. Title: "Average Monthly Temperature (°C)"
    b. X-axis label: "Month"
    c. Y-axis label: "Temperature (°C)"
    d. Rotate x-axis labels by -30° for clarity.
       Add horizontal grid lines.
    e. Enable autoscaling on Y-axis.

F. Color:
   Use a single consistent color (e.g., orange or skyblue) for all bars.

G. Output
   Save your plot as a PDF named avg_temperature_histogram.pdf in your working directory.

H. Expected Output
    a. A clean and professional-looking histogram showing:
    b. 12 bars (one per month)
    c. Properly labeled axes and title
    d. Rotated month labels
    e. Even spacing and grid line

Submit a Gnuplot script named task1.gp that generates annual_sales.pdf according to the requirements above with .dat file.

Task 2: Multi-Bar Histogram of Annual Sales by Region and Company [10 Marks]

Use Gnuplot to generate a multi-bar clustered histogram that compares the annual sales (in millions of dollars) of four companies across six regions.

Dataset File (Input: annual_sales.dat)

Region	CompanyA	CompanyB	CompanyC	CompanyD
North	120	150	130	140
South	100	95	110	120
East	140	160	135	150
West	90	105	100	95
Central	130	120	125	135
Overseas	200	180	190	175

Column 1 → Region (categorical x-axis: North, South, East, West, Central, Overseas)
Columns 2–5 → Annual sales figures (in millions) for Company A–D

Plotting Requirements

1. Output file name: annual_sales.pdf
2. Terminal: pdfcairo
3. Page size: 20 cm × 14 cm
4. Font: "Times New Roman", size 12
5. Plot type: multi-bar clustered histogram
6. Style Settings
    a. Histogram style: clustered with a gap of 1.
    b. Bar fill: solid, 70% opacity.
    c. Box width: 0.8 relative.

    d. Colors and line styles:

        - Company A → Blue, solid

        - Company B → Red, dashed
        - Company C → Green, dotted
        - Company D → Orange, dash-dot
1. Axis & Labels
    a. Title: "Annual Sales Performance by Region (4 Companies)"
    b. X-axis: "Region"
    c. Y-axis: "Sales (in Million $)"
    d. Rotate x-axis labels by -30° for readability.
    e. Add horizontal grid lines.
    f. Autoscale Y-axis.
2. Legend
    a. Position: outside top right, horizontal.

Expected Output

A PDF showing:

1. 6 clusters of bars (one for each region).
2. Each cluster has 4 bars representing Company A–D.
3. Clear color-coded distinction between companies.
4. A descriptive title, axis labels, rotated region labels, and a top legend.
5. A visually clean, professional comparison of company performance across regions.

Deliverable

Submit a Gnuplot script named task2.gp that generates annual_sales.pdf according to the requirements above with the .dat file.

Task 3: You are provided with a file **monsoon_rainfall.dat** containing rainfall (in mm) for 5 regions over 4 months (June–September). The file has three columns:

[10 Marks]

Region	Month	Rainfall
1	6	120
1	7	150
1	8	200
1	9	250
2	6	100
2	7	180
2	8	220
2	9	300
3	6	80
3	7	160
3	8	210
3	9	290
4	6	50
4	7	140
4	8	190
4	9	270
5	6	70
5	7	130
5	8	180
5	9	260

Tasks:

A. Data Preparation:
   a. Ensure the data is structured with regions as blocks and months in increasing order.

b. Use month numbers: 6 = June, 7 = July, 8 = August, 9 = September.

B. Heatmap Plotting in Gnuplot:

a. Plot a 2D heatmap of rainfall intensity.
b. X-axis: Region (labeled as Region 1, Region 2, …)
c. Y-axis: Month (labeled as June, July, August, September)
d. Use color intensity to represent rainfall (low → light colors, high → dark colors).
e. Smooth the heatmap by interpolating additional points
f. Add title as "Monsoon Rainfall Intensity (June–September)", axis labels, and color bar scale.

C. Output:

a. Save the heatmap as monsoon_heatmap.pdf.

Hints:

- Use set xtics and set ytics to label axes with region and month names.

Expected Outcome:
A clearly labeled 2D heatmap showing variations in rainfall intensity across regions and months.

Deliverable

Submit a Gnuplot script named task4.gp that generates monsoon_heatmap.pdf according to the requirements above with the .dat file.